

TEAM ANVESHAK

Mechanical
Application

INSTRUCTIONS:

Please read the instructions below carefully.

(1) Attempt as many questions as you can. Do not worry if you are stuck in any of the questions. Your approach is more important than the final answer.

(2) Answer each question in a new page.

(3) Calculations and drawings/illustrations can be added to the submission pdf as scanned pictures. You may attach any of the models/simulations you created in a google drive link wherever necessary. Make sure you give access to apps.anveshak@gmail.com

(4) Please merge the Questionnaire pdf prior to your answers document and submit it as a PDF in the submission link at http://join.teamanveshakiitm.com/

(5) Attach all the references at the very end of the document.

(6) The filename must follow the format <Roll Number>_MECH.pdf [in Block Letters]. For example, XY24B123_MECH.pdf

(7) If you have any queries regarding the questions, post it on the WhatsApp group or contact: (through WhatsApp)

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QUESTION 1

A mechanical linkage is an assembly of rigid bodies (links) connected by joints to manage forces and control motion. These systems transmit, amplify, or transform movement such as converting rotary motion to linear motion. One such mechanism is as shown, solve for the following using the given data.

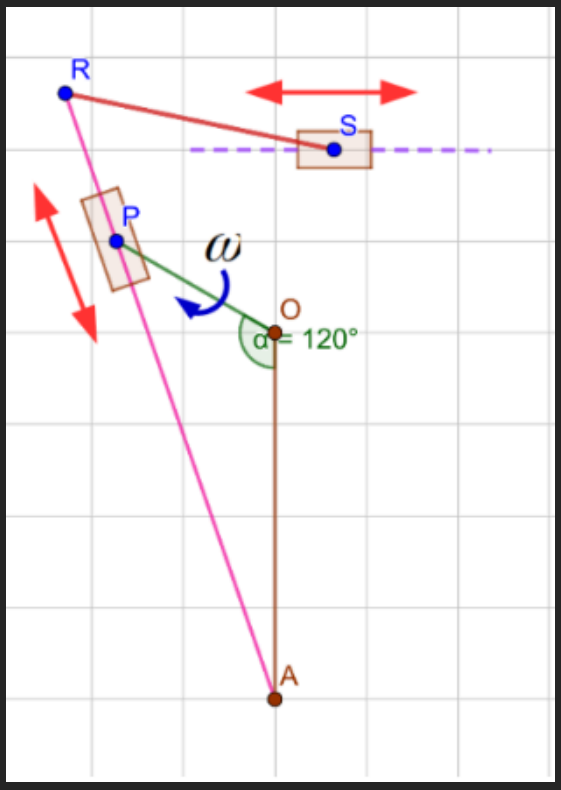


FIGURE 1: SLIDER-CRANK MECHANISM

1.1 In Figure 1, link OA is fixed and is 400mm in length. The link OP is 200mm in length, while AR = 700mm and RS = 300mm. The vertical height from O to S is 200mm. The angle AOP at this instant is 120 degrees. The angular velocity of link OP is in the clock-wise direction and has a magnitude of 7π rad/s. Determine:

1. Acceleration of 'S'
2. Angular acceleration of link RS

1.2 Define degree of freedom. Identify and label the degrees of freedom in given Figure 2 as roll, pitch or yaw.

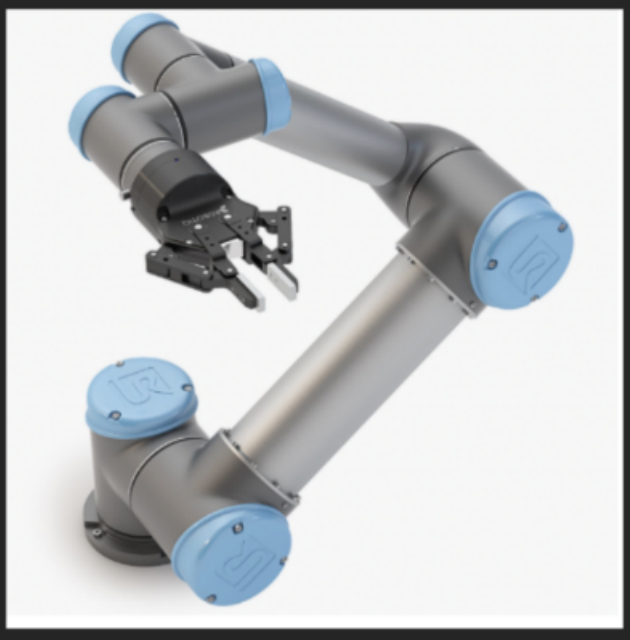


FIGURE 2: MANIPULATOR ARM

QUESTION 2

Consider a four-wheel rover of mass m performing differential (skid-steer) rotation in place on a flat horizontal surface with coefficient of friction μ. The rover has a track width W, wheelbase L, radius of wheel r, height of COM h and identical wheel–motor units. Assume the normal load is equally distributed among all wheels.

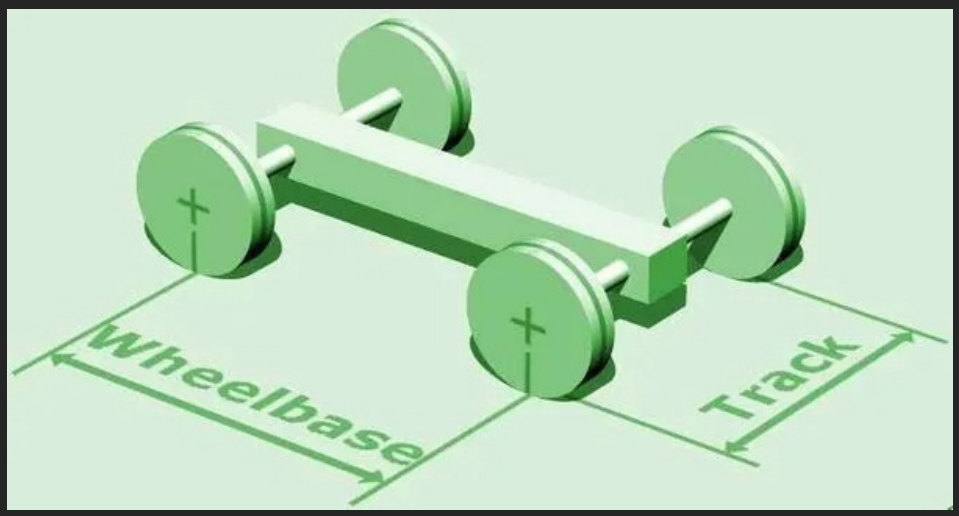


FIGURE 3: THE NOMENCLATURE

2.1 Derive the expression for the minimum motor torque per wheel required to achieve steady rotation in place in full lateral slip regime. How do the given geometric parameters affect the required torque?

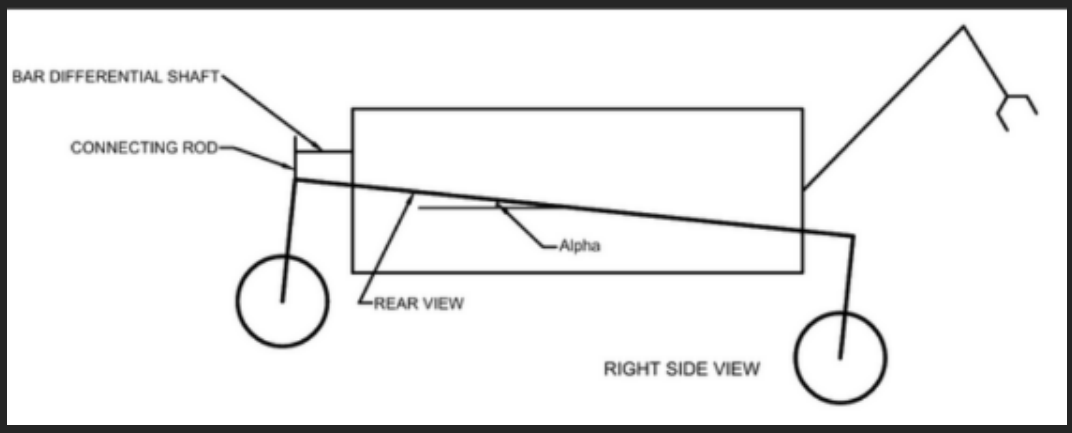
2.2 Now the rover is turning while driving forward using differential drive along a circular arc of radius R. Which wheel(inner or the outer) will tend to lateral slip first? Knowing this determine the maximum velocity the rover can move around the circle without causing lateral slip. Note that weight is not equally distributed on the wheels due to lateral acceleration. Assume a linear tire-ground model
($F_{friction} = C_\alpha * \alpha$, where C_α is cornering stiffness and α is slip angle).

QUESTION 3

The bar differential plays a key role in maintaining stability for the rover during traversal. It features a main differential bar, pivoted at one or two points on the rover chassis. A connecting rod on each side links the differential bar to the rear rocker links, with two ball joints on each side used to connect the rods.

You can refer to the first part of the video (here) for visualizing the functioning of single pivot bar differential.

3.1 Suppose we are using a rear single pivot point bar differential in our rover. Alpha is the angle made by the rocker links with respect to the horizontal axis and beta is the angle made by the differential with respect to its horizontal axis. Find the relationship between them. Take the origin as the coupler at the left side of the rear end (refer Figure 5). Take suitable assumptions wherever required.



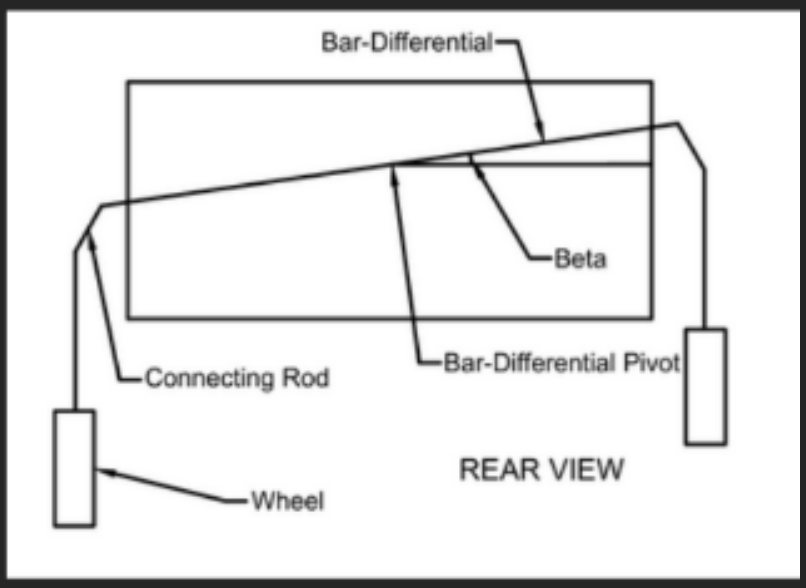


FIGURE 4: REPRESENTATION OF THE SYSTEM

3.2 Even though it is optimal to put the chassis coupler at the exact COM, we can't precisely calculate the COM due to various reasons. The couplers are attached at 25 cm from the backside of the chassis. Assume the COM is 5 cm backward in the x direction from the coupler. There are 2 wedges angled at 30 degrees placed sidewise as shown in the image below (y and z coordinates of COM you can take as the middle of the respective lengths chassis). Take the origin as the coupler at the left side of the rear end.

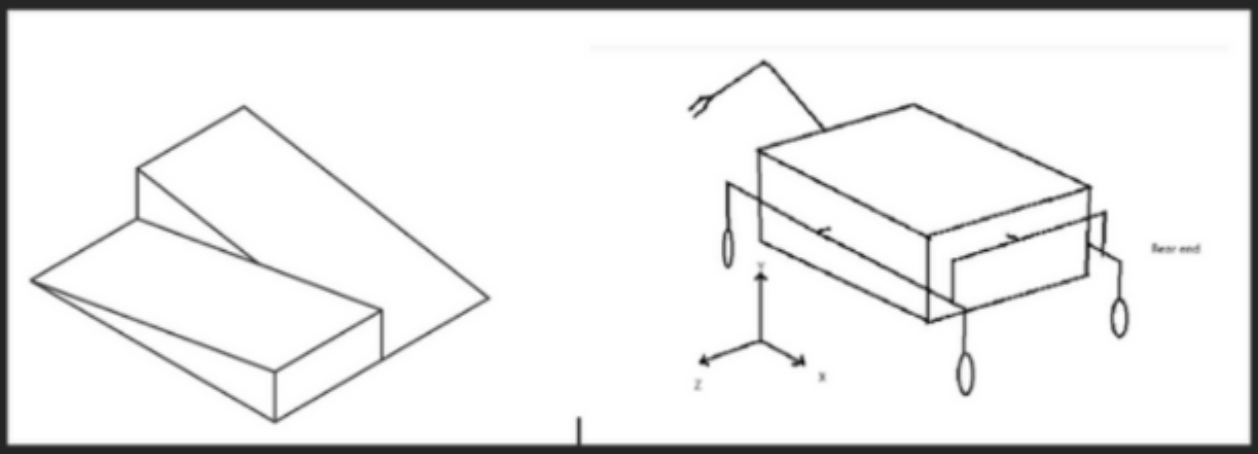


FIGURE 5: THE SETUP AND CO-ORDINATE AXES

The rover is put on the wedges with the same orientation as in the image such that each pair of wheel on each side is resting on respective wedge. Find the instantaneous reaction force vector on,

1. Bar differential pivot
2. At both ends of main differential bar

You can take suitable assumptions if needed.

- Mass of the rover = 50 kg
- g = 10 m/s^2
- Dimension of chassis = 50 cm x 50 cm x 40 cm
- Rear and front rocker link length is equal and each length = 35 cm
- Connecting rod length = 20 cm
- Bar differential length = 60 cm

3.3 The rover moves along a banked surface that is part of a circular track of radius R and a superelevation (inclination) of α. The mass of the rover is M Kg and is incorporated with a double rocker suspension system as shown in the figure. Assume all the rocker links are rigid and have a rectangular cross section. Find out the value (and where) of maximum stress within the link. The length of link is 1.5m, rocker angle is 120 degrees and the dimension of the chassis 2m x 2m. Make necessary assumptions and justify them. Ignore shear stresses.

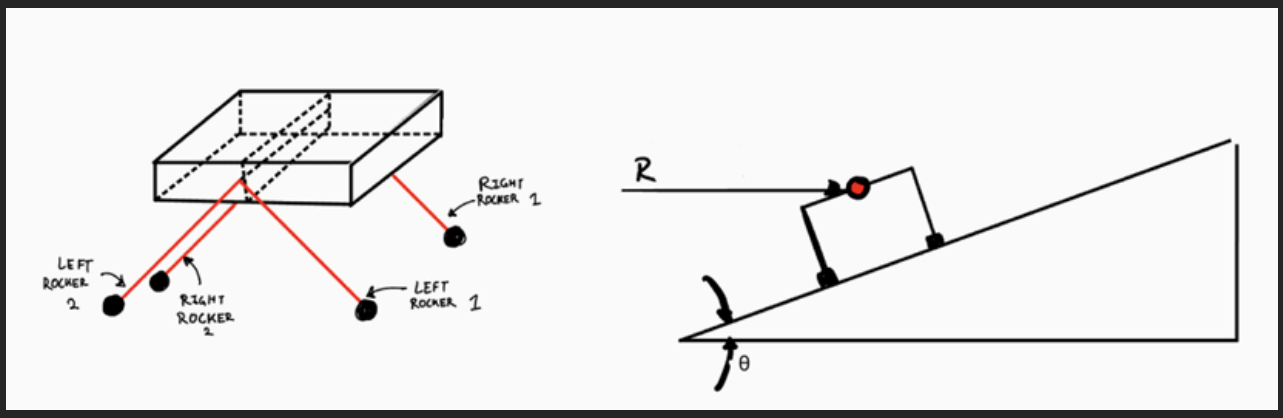


FIGURE 6: VISUAL REPRESENTATION

Some important things to keep in mind:

1. The red dot reppresents the centre of mass (CoM).
2. The above setup is completely different from the ones used in previous two subquestions.
3. Given setup uses rocker mechanism (research more on the same) instead of the parallel links, the ones used previously.
4. The rover is facing the screen on the banked road and hence the CoM position is as shown.

QUESTION 4

4.1 There is a bogie of fixed width 'b' and wheels of radius 'r'. Find expressions for y_{min} and the maximum allowable v-angle φ the bogie can have to safely climb an obstacle (say a step) of height h, as given below. To climb the step completely, rear wheel has to come in contact with it. Hence, the rear link shouldn't hit the step before the wheel reaches it. (Assume there is no toppling involved)

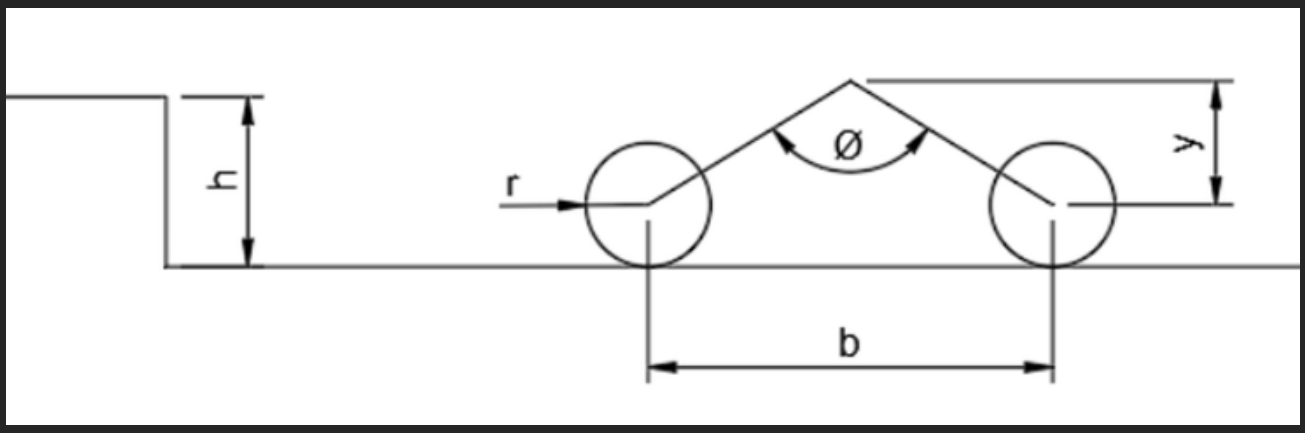


FIGURE 7: NOMENCLATURE AND REPERESENTATION

Bonus: Try simulating this motion in any software.

4.2 In regular vehicles, circular wheels naturally provide a smooth ride on flat roads. But if we have triangular wheels, will they work on our current roads for a smooth ride?

1. What does a 'smooth ride' mean in terms of rolling motion?
2. Find the equation for the road shape needed for a smooth ride with equilateral triangular wheels. Explain the math behind this equation.
3. If we use these triangular wheels on the road (obtained above), will there be any problem or not? If yes, than why?

QUESTION 5

5.1 The manipulator of the rover requires large amounts of torque to be produced by the motors in order to be able to lift heavy objects. To this end, we use gearboxes at the shoulder, elbow, and base rotation joints with high gear reduction ratios. Different gearing systems can be used for this purpose, such as cycloidal, worm gear, and planetary gearing.

Given below are specifications for different gearing systems. Research and explain how the gear reduction ratios for each of these systems have been arrived at, and calculate the ratios for each of the five systems:

1. A worm with 2 starts, and a worm wheel of 20 teeth. (Reference Video)
2. A planetary gear system with a sun gear of 15 teeth, three planet of 30 teeth each, and a ring gear of 75 teeth, where
    (a) the ring gear is fixed
    (b) the sun gear is fixed
    (c) the carrier is fixed

    (Reference Video)

5.2 A pair of parallel helical gears consists of a 20 teeth pinion meshing with a 100 teeth gear. The pinion rotates at 720 rpm. The normal pressure angle is 20° while the helix angle is 25°. The face width is 40 mm and the normal module is 4 mm. The pinion as well as the gear is made of steel 40C8 (S_{ut} = 600 N/mm^2) and is heat treated to a surface hardness of 350 BHN. The service factor and the factor of safety are 1.5 and 2.0 respectively. Assume that the velocity factor accounts for the dynamic load.

Calculate the power transmitting capacity of the gears.

Explain any assumptions taken with derivations.



FIGURE 8: HELICAL GEARS

QUESTION 6

There are four rigid body components shown below, each shown in different colours. Shaft D is stationary. The disc A has a hole of radius R_h for the shaft of component B (of radius R_r) to roll in (with slipping). Disc A also has a small hole at its centre for the shaft of C to pass through. Component B can freely rotate about D (about axis 1). Disc A rotates about the shaft of C with angular velocity ω_1 (about axis 2). Component C rotates about D with angular velocity ω_2 (about axis 1). Therefore, A has a revolutionary motion about axis 1 and a rotational motion about axis 2. As a result of A's motion, component B will be pushed by the surface of the hole of A. The centers of disc A, shaft of B, shaft of C and D are along the same line initially and the distance between centres of C and D is $R_h - R_r$.

6.1 If the angle by which B has rotated about axis 1 is θ in time t, then find θ as a function of time.

6.2 Is there any condition under which the shaft of B undergoes pure rolling, i.e., rolls without slipping in the hole of A?

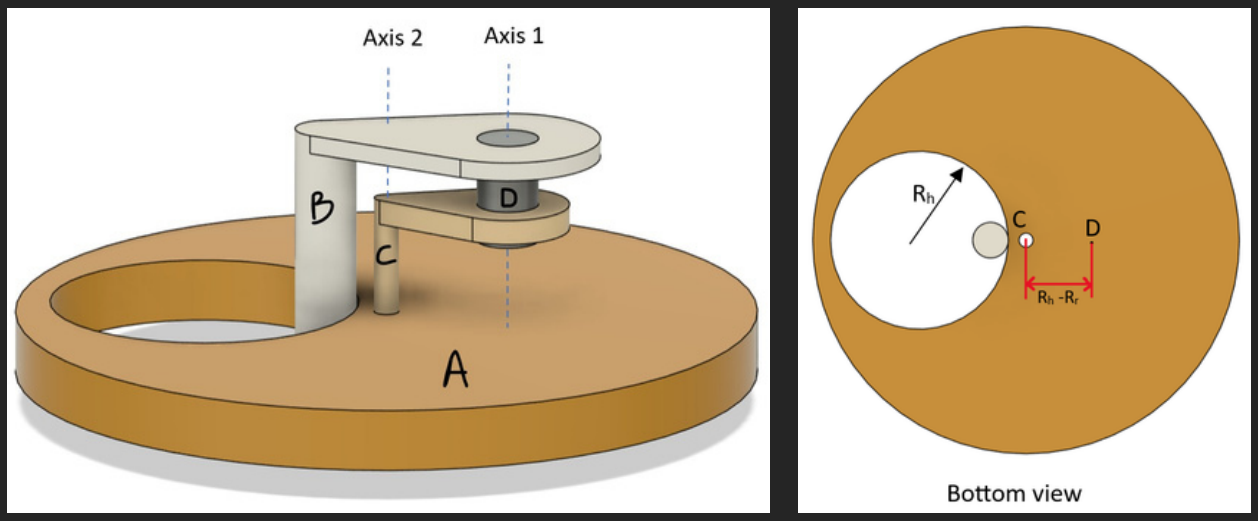


FIGURE 9: THE RIGID BODY SETUP

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